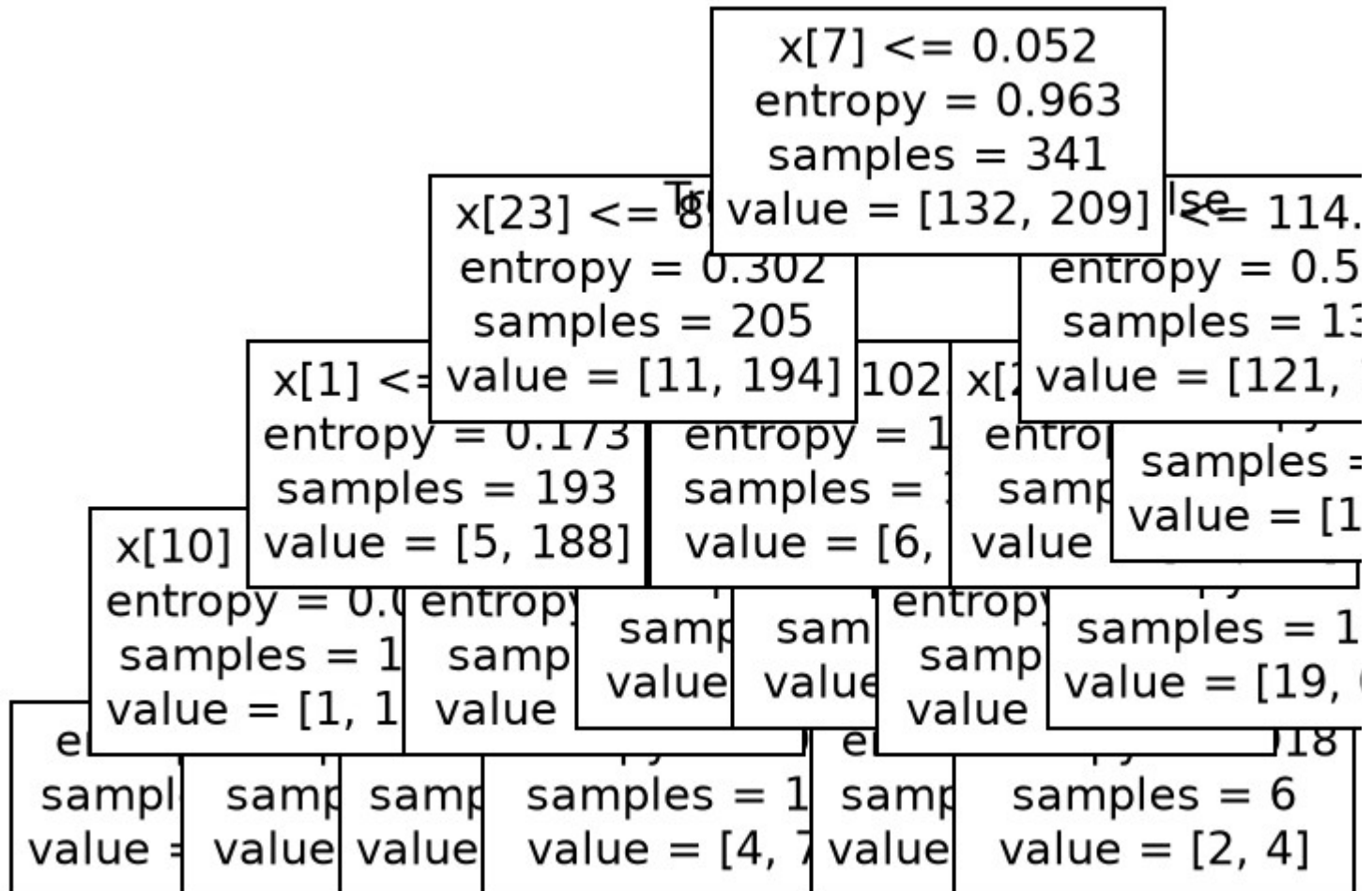
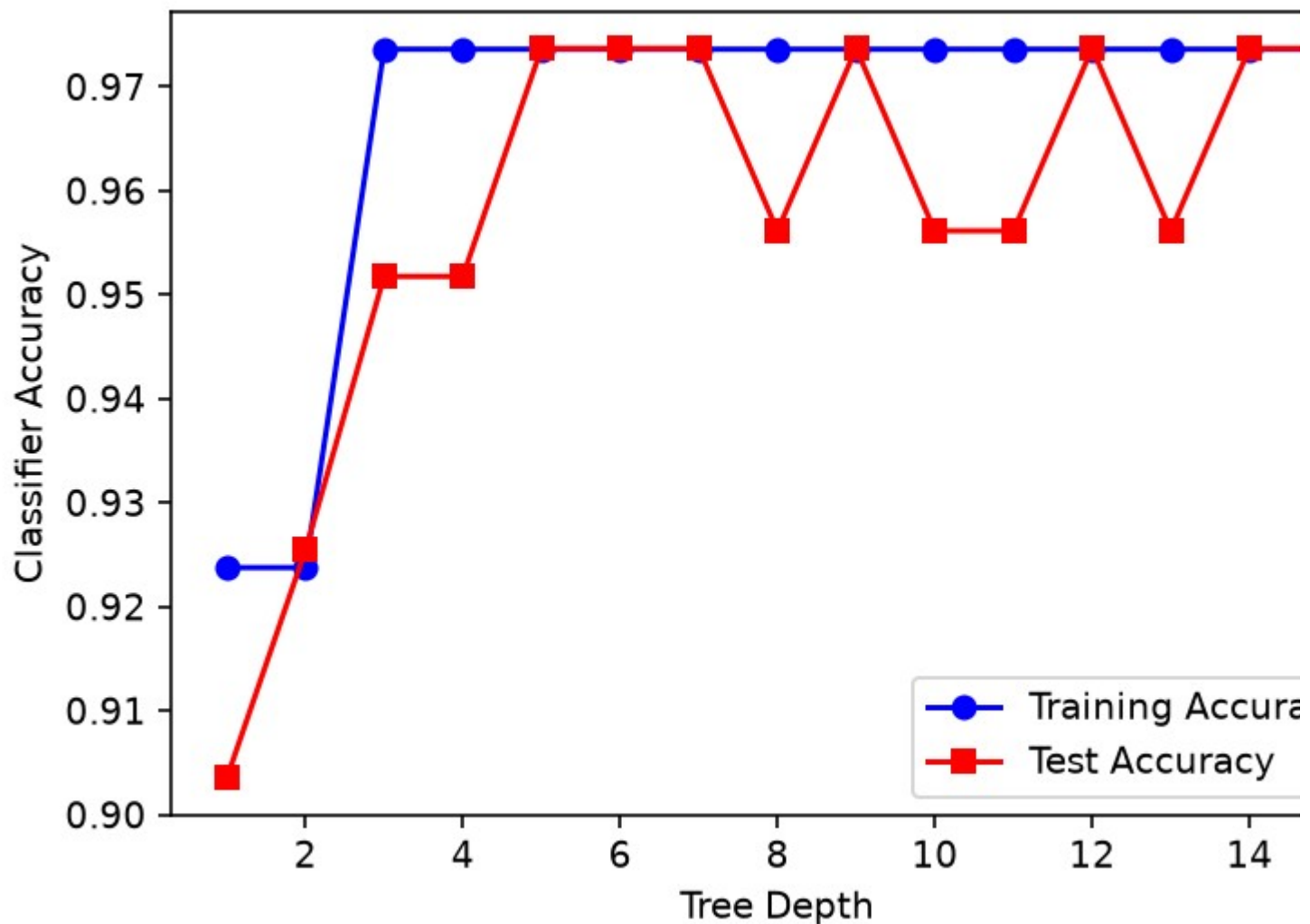


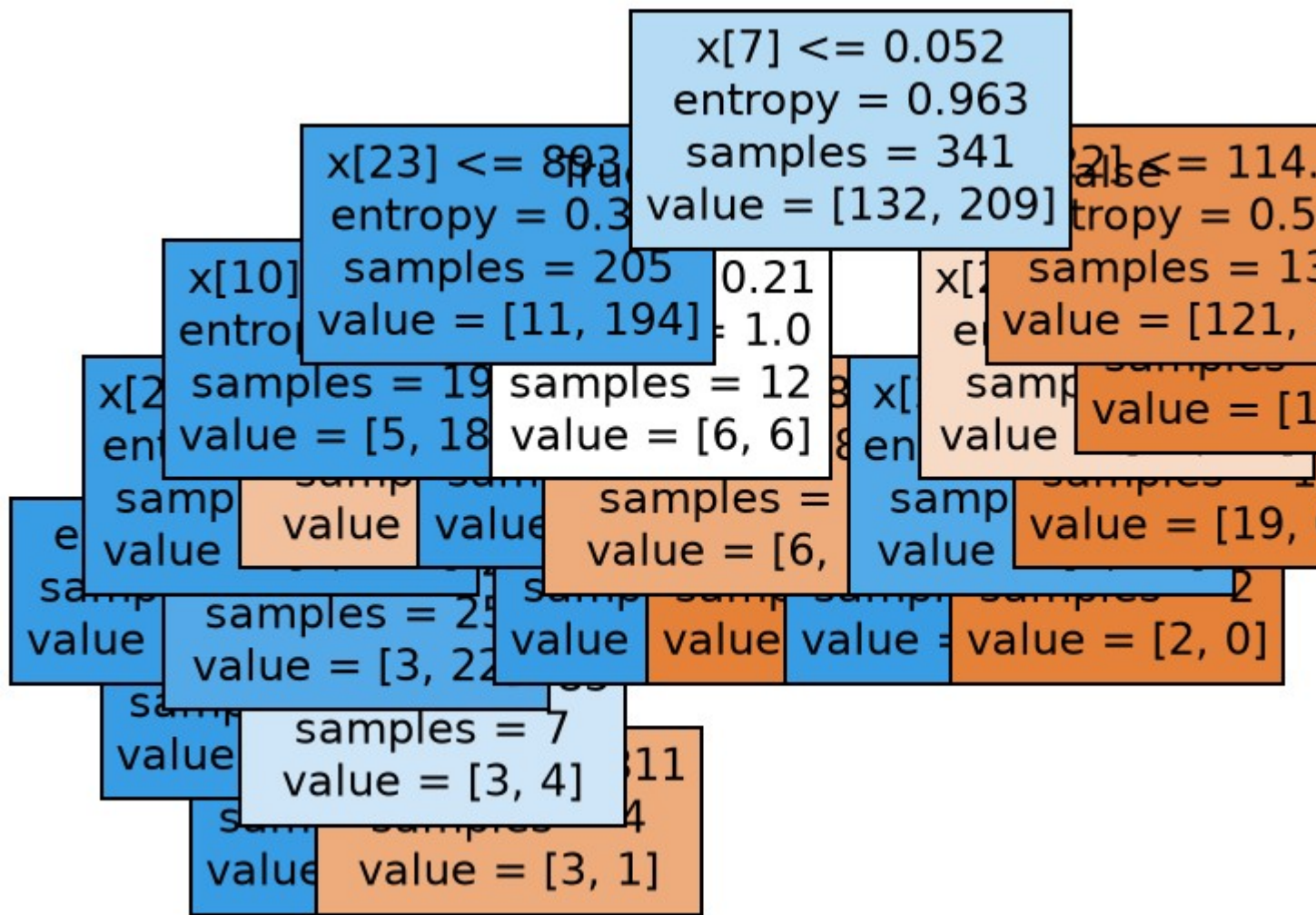
```
In [69]: %runcell -i 0 C:/Projects/torontomu/Assignment1/Assignment1.py
There are 569 instances described by 30 features.
The accuracy of the classifier is 0.956140350877193
```



```
In [70]: %runcell -i 1 C:/Projects/torontomu/Assignment1/Assignment1.py
Out[70]: " \nAccording to the test accuracy, the best model has a maximum depth of
approximately 3, 5, 10, 11, or 13, which tie for the highest test accuracy. \nHowever,
we should not use the test data to select hyperparameters because doing so can overfit
to the test set and lead to an overly optimistic estimate of the model's performance.\n"
```



```
In [71]: %runcell -i 2 C:/Projects/torontomu/Assignment1/Assignment1.py
Fitting 5 folds for each of 75 candidates, totalling 375 fits
Best params: {'max_depth': 12, 'min_samples_split': 7}
Best CV score: 0.9178175618073316
Best estimator: DecisionTreeClassifier(max_depth=12, min_samples_split=7)
The maximum depth of the tree is 12 and the minimum number of samples required to split
a node is 7
Out[71]: '\nSplit the training dataset into 10 folds, and for each fold, train the model
on the other 9 folds and validate it on the current fold. This process is repeated for
all 10 folds, and the results are averaged to provide a more robust estimate of model
performance.\nStratified cross-validation ensures that each fold has approximately the
same percentage of samples of each target class as the complete set, which is
particularly important for imbalanced datasets.\n'
```



In [72]: